

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  struct node
4  {
5      int data;
6      struct node *next;
7  };
8
9  struct node *front = NULL, *rear = NULL;
10
11 void enqueue()
12 {
13     int value;
14     struct node *newnode;
15     newnode = (struct node *) malloc (sizeof(struct node));
16     if (newnode == NULL) {
17         printf("insufficient memory.\n");
18         return;
19     }
20     else {
21         printf("Enter the data: ");
22         scanf("%d",&value);
23         newnode->data = value;
24         newnode->next = NULL;
25         if (front == NULL && rear == NULL) {
26             rear = front = newnode;
27         }
28         else {
29             rear->next = newnode;
30             rear = newnode;
31         }
32         printf("%d is inserted\n", newnode->data);
33     }
34 }
35
36 void dequeue()
37 {
38     struct node *temp;
39     if (front == NULL) {
40         printf("underflow condition arise\n");
41         return;
42     }
43     else {
44         temp = front;
45         printf("%d is deleted\n", temp->data);
46         front = front->next;
47         if (front == NULL)
48             rear = NULL;
49         free(temp);
50     }
51 }
52
53 void display()
54 {
55     struct node *temp;
56     if (front == NULL) {
57         printf("Queue is empty.\n");
58         return;
59     }
60     temp = front;
61     while (temp != NULL) {
62         printf("%d ", temp->data);
63         temp = temp->next;
64     }
65     printf("\n");
66 }

```

```
67
68 int main()
69 {
70     int ch;
71     while (1) {
72         printf("Queue Menu\n");
73         printf("1. Enqueue\n");
74         printf("2. Dequeue\n");
75         printf("3. Display\n");
76         printf("4. Exit\n");
77         printf("Enter your choice (1/2/3/4): ");
78         scanf("%d",&ch);
79         switch (ch)
80         {
81             case 1:
82                 enqueue();
83                 break;
84             case 2:
85                 dequeue();
86                 break;
87             case 3:
88                 display();
89                 break;
90             case 4:
91                 return 0;
92             default:
93                 printf("your entered wrong choice\n");
94         }
95     }
96     return 0;
97 }
98
```